

Graph Conditional Denoising Autoencoder with Random Fourier Features for Climate Adaptation

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1 Synthetic Dataset Design

The synthetic dataset is a two-modality dataset with features designed to resemble climate and sociodemographic features observed in real settings. To mimic a real geographic region, I construct hierarchical spatial regions with two levels, covering an area that roughly corresponds to the continental United States. The fine-level grid contains 5,000 regions, and the coarse-level grid contains 50 regions. Both levels use the projected coordinate system EPSG:5070. This ensures that distance-based calculations are meaningful and accurate when simulating spatial random fields.

The synthetic dataset contains three climate-like features and three sociodemographic-like features. I assume that all features are generated from a shared spatial random field at the coordinate level, together with modality-specific spatial random fields. This structure induces positive correlation between the two modalities. In addition, to mimic a realistic setting, we further assume that the climate-specific features vary smoothly over space, while the sociodemographic-specific features have sharper local variation. The shared spatial field is generated using a basis-function approximation to a Gaussian process (GP) with a Matérn kernel and smoothness parameter ($\nu = 1.5$) (Banerjee et al., 2008), shown as

$$\phi_{ik} = \left(1 + \frac{\sqrt{3} \|s_i - \kappa_k\|_2}{\ell}\right) \exp\left(-\frac{\sqrt{3} \|s_i - \kappa_k\|_2}{\ell}\right), \quad \tilde{x}_i = \sum_{k=1}^K \phi_{ik} w_k, \quad (1)$$

where s_i is the i th-coordinate, κ_k is the coordinate of k th-knot, K is the number of knots, ϕ_{ik} is the basis function between i th-location and k th-knot, and $w_k \sim \mathcal{N}(0, 1)$ is the random basis weight. The length-scale parameter ℓ is set to 500,000 meters to represent mid-range spatial dependence.

The climate-specific spatial fields are generated using the same basis-function approximation in (1), but with longer length-scale parameters of 800,000, 1,000,000, and 1,200,000 meters for the three features. On the other hand, the sociodemographic-specific spatial fields are generated using a conditional autoregressive model (CAR) on the fine-level grid, which reflects the neighborhood dependence commonly observed in area-level survey data (Besag, 1974; Besag and Kooperberg, 1995). The CAR model is defined as

$$\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \sigma^2(\mathbf{D} - \rho\mathbf{A})^{-1}),$$

where σ is the marginal standard deviation, $\mathbf{A}$ and $\mathbf{D}$ are adjacency matrix and degree matrix of fine-level grid respectively. The spatial autocorrelation ρ is set to (0.85, 0.88, 0.9) for three sociodemographic-specific spatial fields. Finally, all features are defined as linear combinations of the shared spatial field and their corresponding modality-specific spatial fields, with the shared component contributing a proportion of 0.3.

In addition, I incorporate two forms of structured heterogeneity and missingness. First, because sociodemographic variables often vary systematically across states due to differences in policy and administration, I add a coarse-level random effect to the sociodemographic modality. Second, I introduce modality-specific block missingness. For the climate-like features, this can represent missing satellite observations caused by cloud cover. For the sociodemographic-like features, it can represent suppression of sensitive survey data in nearby low-income regions.

For this project, three target variables are constructed. The first two are defined as linear combinations of both climate-like and sociodemographic-like features. The third target is defined as a nonlinear function of all features, which allows us to evaluate whether the proposed model captures nonlinear relationships. Lastly, idiosyncratic noise is added to each target to represent measurement error. The plot of the final dataset can be found in the README file.

2 Self-Supervised Model

The small-scale foundation model developed in this project is designed to capture spatial dependence in area-level data while also representing continuous spatial structure, particularly long-range spatial dependence. This is achieved by combining a graph neural network with spatial random Fourier features. I denote this model as graph conditional denoising autoencoder (GCDAE) with random Fourier features. In addition, this foundation model is able to handle multi-modality and potential hierarchical group effect in the dataset.

Random Fourier features (RFFs) provide an efficient way to encode position information and spatial dependence, including both local and long-range structure Rahimi and Recht (2007); Daw and Wikle (2023). For each i th location, the RFF can be constructed as

$$\mathbf{r}_i = \sqrt{\frac{2}{K}} \left[\cos(\boldsymbol{\omega}_1^\top \tilde{\mathbf{s}}_i + b_1), \dots, \cos(\boldsymbol{\omega}_K^\top \tilde{\mathbf{s}}_i + b_K) \right],$$

where $\tilde{\mathbf{s}}_i$ is the standardized coordinate of i th location, $\boldsymbol{\omega}_k$ is the k th random frequency, b_k is the k th random phase, and K is the number of RFFs. In particular, climate, and sociodemographic features may contain both local spatial variation and long-range dependence. For this reason, $\boldsymbol{\omega}_k$ are sampled from

$$\boldsymbol{\omega}_k \sim \mathcal{N}\left(\mathbf{0}, \frac{3}{\ell^2} \mathbf{I}\right),$$

with a wide range of length scales, $\ell \in \{0.1, 0.3, 0.5, 0.8, 1.2\}$, to represent spatial dependence at several ranges.

The overall structure of the model follows a conditional generative framework, as in Sohn et al. (2015). The encoder takes the input features together with two sets of conditioning variables, the one-hot encoded coarse-region identifier and random Fourier features of the fine-grid centroids. These inputs are first passed through a dropout layer with a dropout rate of 0.2. The perturbed inputs are then fed into a two-layer graph convolutional network (GCN). The two GCN layers allow the model to aggregate information from two-hop neighborhoods, which is sufficient to represent the local neighborhood structure in the sociodemographic features. The GCN output is then mapped by a linear layer to produce the learned embedding. Finally, the embedding is partitioned into two segments, one for the climate-like features and one for the sociodemographic-like features.

The decoder consists of two modality-specific networks, one for each modality. Each network takes the embedding segment corresponding to its modality, together with the same conditioning variables used in the encoder, and passes them through a GCN with an architecture symmetric to that of the encoder. At the output layer, each decoder reconstructs its corresponding modality-specific features. The training objective is the mean squared error between the reconstructed features and the original unperturbed features. Entries that are missing in the original data are masked in the loss function to prevent them from distorting the learning objective. Parameters are tuned to minimize the reconstruction error of the input features. Since the self-supervised stage is an open-training problem based on unlabeled features, the concern about test leakage does not apply. Table 1 displays the reconstruction performance of GCDAE and GCDAE with RFF (GCDAE-RFF). It shows that adding RFF substantially improves reconstruction accuracy.

Table 1: Reconstruction performance of GCDAE with and without RFF.

Model	Reconstruction RMSE
GCDAE	0.0518
GCDAE-RFF	0.0118

The model design and tuning were strongly influenced by the available computing environment. Because GPU access was limited during the summer, I trained the GCDAE on my laptop, with an AMD Ryzen 5 7640U CPU. As a result, I was not able to search over the full parameter space. For example, deeper GCNs could capture longer-range dependence through additional graph hops, but they would be computationally expensive. The final GCDAE model uses a hidden dimension of 512, a latent dimension of 256, and a dropout rate of 0.2.

3 Downstream Interpolation Evaluation

The downstream analysis evaluates interpolation performance on the three simulated targets. I fit a ridge regression model using the embeddings learned by GCDAE-RFF, which is simply referred to as GCDAE-RFF in the following discussion. I compare it with four baseline models: inverse-distance weighting (IDW), kNN regression in coordinate space, kNN regression using spatial empirical orthogonal functions (kNN-EOFs), and a graph convolutional network (GCN) using coordinates and EOFs as inputs. Spatial EOFs provide a simple way to capture low-frequency spatial structure, including long-range dependence, and are computed from the eigenvectors of the fine-level grid adjacency matrix (Porter et al., 2014). Table 2 illustrates that GCDAE-RFF consistently outperforms all baseline methods. I also note that the GCN baseline may not be fully optimized because of limited computation resources.

Table 2: Interpolation performance on the three simulated targets.

Model	y_1	y_2	y_3	Total
IDW	0.1588	0.2869	0.1266	0.1907
kNN	0.1567	0.2800	0.1262	0.1876
kNN-EOF	0.1580	0.2819	0.1275	0.1891
GCN	0.1551	0.2815	0.1312	0.1893
GCDAE-RFF	0.1481	0.2594	0.1235	0.1770

4 Discussion

In this project, I show that a small-scale foundation model conditioned on physical and spatial assumptions can improve downstream interpolation performance. In practice, several issues remain for applying foundation models to real-world datasets. First, real-world datasets are often spatio-temporal, so using RFFs for spatial structure alone may not be sufficient. A natural extension is to incorporate physical mechanisms directly into the network transition function or the training objective. For example, given a partial differential equation (including those defined on graphs), one could derive a transition function using the Euler method and implement it as a spatially aware recurrent neural network.

Real spatial datasets often contain multiple geospatial hierarchies. To generate embeddings across these levels, we can define models with input layers corresponding to each spatial scale. For example, within a foundation model, state-level representations can be aggregated into county-level representations using a simple pooling layer or an attention layer. The county-level input can then combine the aggregated state-level embedding with county-level features to learn a more informative county-level embedding.

Another issue is that sociodemographic data, especially survey data, are often aggregates of sampled units within each region. As a result, sampling variance is commonly present in these datasets. Foundation models could be further improved by incorporating the reported sampling variance into the objective function.

For causal inference tasks, the learned model can be fine-tuned using the features and treatment variable from the downstream dataset. The treatment variable can be mapped to an additional embedding segment with a sigmoid activation, which can be interpreted as an estimated propensity score. This propensity score can then be used for matching, weighting, or incorporated jointly with the learned embeddings in the downstream outcome model.

5 AI Assistance and Disclosure

For coding, I use ChatGPT to search for tedious functions, such as plotting utilities and NumPy functions. But I do not rely on AI for model design. For writing, I use ChatGPT mainly for proofreading.

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